

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE NAME:** CSA16 **COURSE CODE:** Data Warehousing and Data Mining

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### COURSE OUTCOME COVERED

**CO3:** Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)  
Assessment Tool Weightage: 50%

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**QUESTIONS: 5**

**Total Marks:50**

Annexure A

Experiments

| <b>Criteria</b>  | Understanding of Concepts<br><b>(2.5)</b> | Experimental Design <b>(3)</b> | Use of Tools<br><b>(2)</b> | Analysis of Results <b>(1.5)</b> | Documentation<br><b>(1)</b> | <b>Total(10)</b> |
|------------------|-------------------------------------------|--------------------------------|----------------------------|----------------------------------|-----------------------------|------------------|
| <b>Weightage</b> | 25%                                       | 30%                            | 20%                        | 15%                              | 10%                         | 100%             |
| <b>Q1</b>        |                                           |                                |                            |                                  |                             |                  |
| <b>Q2</b>        |                                           |                                |                            |                                  |                             |                  |
| <b>Q3</b>        |                                           |                                |                            |                                  |                             |                  |
| <b>Q4</b>        |                                           |                                |                            |                                  |                             |                  |
| <b>Q5</b>        |                                           |                                |                            |                                  |                             |                  |

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### **Code of Conduct:**

I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

**RUBRICS FOR CALCULATION:**

| <b>Criteria</b>           | <b>Weightage</b> | <b>Level 4 – Exemplary</b>                                                                    | <b>Level 3 – Proficient</b>                                 | <b>Level 2 – Developing</b>                         | <b>Level 1 – Beginning</b>                       |
|---------------------------|------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------|
| Understanding of Concepts | 25               | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding; unable to integrate concepts |
| Experimental Design       | 30               | Well-planned, systematic implementation with optimal solution                                 | Correct implementation with minor issues                    | Basic implementation with errors or inefficiencies  | Incorrect or incomplete implementation           |
| Use of Tools              | 20               | Effective and advanced use of tools/technologies with proper configuration                    | Appropriate use of tools with correct execution             | Limited or inconsistent use of tools                | Incorrect or minimal use of tools                |
| Analysis of Results       | 15               | Insightful analysis with accurate interpretation and validation of results                    | Good analysis with reasonable interpretation                | Basic analysis with limited insights                | Poor or incorrect analysis of results            |
| Documentation             | 10               | Clear, well-structured documentation with diagrams, code, and results                         | Organized documentation with necessary details              | Basic documentation with missing details            | Poor or incomplete documentation                 |

## Annexure C

### Experiments :

Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

| Age of child | Photograph: |    |    |
|--------------|-------------|----|----|
|              | A           | B  | C  |
| 5-6 years:   | 18          | 22 | 20 |
| 7-8 years:   | 2           | 28 | 40 |
| 9-10 years:  | 20          | 10 | 40 |

Use cov() to calculate the sample covariance between B and C.

Use another call to cov() to calculate the sample covariance matrix for the preferences.

Use cor() to calculate the sample correlation between B and C.

Use another call to cor() to calculate the sample correlation matrix for the preferences.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

### Player ID Player Name Points Scored

|     |          |    |
|-----|----------|----|
| P1  | Player A | 12 |
| P2  | Player B | 15 |
| P3  | Player C | 14 |
| P4  | Player D | 10 |
| P5  | Player E | 18 |
| P6  | Player F | 20 |
| P7  | Player G | 22 |
| P8  | Player H | 13 |
| P9  | Player I | 11 |
| P10 | Player J | 35 |
| P11 | Player K | 16 |

|     |          |    |
|-----|----------|----|
| P12 | Player L | 17 |
| P13 | Player M | 19 |
| P14 | Player N | 21 |
| P15 | Player O | 23 |

3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**.  
(10 Marks)

Consider the Data Set

young,high,employed,good,yes  
 young,high,self-employed,average,yes  
 middle,high,employed,good,yes  
 old,medium,employed,good,yes  
 old,low,unemployed,poor,no  
 old,low,self-employed,average,no  
 middle,low,unemployed,poor,no  
 young,medium,employed,average,yes  
 young,low,unemployed,poor,no  
 old,medium,self-employed,good,yes  
 middle,medium,employed,average,yes  
 middle,high,self-employed,good,yes  
 old,medium,unemployed,poor,no  
 young,medium,self-employed,average,yes  
 middle,low,employed,poor,no

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50

(i) Find which class had scored higher mean, median and range.

(ii) Plot above in boxplot and give the inferences

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

|             |      |      |      |      |      |      |      |      |      |
|-------------|------|------|------|------|------|------|------|------|------|
| <i>age</i>  | 23   | 23   | 27   | 27   | 39   | 41   | 47   | 49   | 50   |
| <i>%fat</i> | 9.5  | 26.5 | 7.8  | 17.8 | 31.4 | 25.9 | 27.4 | 27.2 | 31.2 |
| <i>age</i>  | 52   | 54   | 54   | 56   | 57   | 58   | 58   | 60   | 61   |
| <i>%fat</i> | 34.6 | 42.5 | 28.8 | 33.4 | 30.2 | 34.1 | 32.9 | 41.2 | 35.7 |

- (a) Calculate the mean, median, and standard deviation of *age* and *%fat*.
- (b) Draw the boxplots for *age* and *%fat*.
- (c) Draw a *scatter plot* and a *q-q plot* based on these two variables.

| age    | income | student | credit rating | buys-computer |
|--------|--------|---------|---------------|---------------|
| <=30   | high   | no      | fair          | no            |
| <=30   | high   | no      | excellent     | no            |
| 31--40 | high   | no      | fair          | yes           |
| >40    | medium | no      | fair          | yes           |
| >40    | low    | yes     | fair          | yes           |
| >40    | low    | yes     | excellent     | no            |
| 31--40 | low    | yes     | excellent     | yes           |
| <=30   | medium | no      | fair          | no            |
| <=30   | low    | yes     | fair          | yes           |
| >40    | medium | yes     | fair          | yes           |
| <=30   | medium | yes     | excellent     | yes           |
| 31--40 | medium | no      | excellent     | yes           |
| 31--40 | high   | yes     | fair          | yes           |
| >40    | medium | no      | excellent     | no            |

01

$$\Rightarrow \text{info}(D) = -\sum_{i=1}^m p_i \log_2(p_i)$$

$$\Rightarrow \text{info}_A(D) = \sum_{s=1}^v \frac{|D_s|}{|D|} \times \text{info}(D_s)$$

$$\Rightarrow \text{Gain}(A) = \text{info}(D) - \text{info}_A(D)$$

\* For Attribute (Age):

| age    | class |   | I(p/N) |
|--------|-------|---|--------|
|        | Y     | N |        |
| <=30   | 2     | 3 | 0.971  |
| 31--40 | 4     | 0 | 0      |
| >40    | 3     | 2 | 0.971  |
|        | 9     | 5 | 0.940  |

$\Rightarrow$  c lass  $P \rightarrow$  Those who buys computer



class N  $\rightarrow$  those who won't buy computer

$$\begin{aligned} \text{(i)} \quad I(p, N) &\Rightarrow I(2, 3) \Rightarrow -\frac{2}{5} \log_2 \left( \frac{2}{5} \right) - \frac{3}{5} \left( \log_2 \left( \frac{3}{5} \right) \right) \\ &\Rightarrow -\frac{2}{5} (-1.322) - \frac{3}{5} (-0.737) \\ &\Rightarrow 0.971 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad I(p, N) &\Rightarrow I(4, 0) \Rightarrow -\frac{4}{4} \log_2 \left( \frac{4}{4} \right) - \frac{0}{4} \log_2 \left( \frac{0}{4} \right) \\ &\Rightarrow -1 \log_2 (1) - 0 \log_2 (0) \\ &\Rightarrow 0 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad I(p, N) &\Rightarrow I(3, 2) \Rightarrow -\frac{3}{5} \log_2 \left( \frac{3}{5} \right) - \frac{2}{5} \log_2 \left( \frac{2}{5} \right) \\ &\Rightarrow -\frac{3}{5} (-0.737) - \frac{2}{5} (-1.322) \\ &\Rightarrow 0.971 \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad I(p, N) &\Rightarrow I(9, 5) \Rightarrow -\frac{9}{14} \log_2 \left( \frac{9}{14} \right) - \frac{5}{14} \log_2 \left( \frac{5}{14} \right) \\ &\Rightarrow -\frac{9}{14} (-0.6374) - \frac{5}{14} (-1.4854) \\ &\Rightarrow 0.940 \end{aligned}$$

$$* \text{Info Age}(D) = \sum_{j=1}^4 \frac{|D_j|}{|D|} \times \text{Info}(D_j)$$

|   |   |    |
|---|---|----|
| 2 | 3 | 5  |
| 3 | 0 | 4  |
| 3 | 2 | 5  |
|   |   | 14 |

$$\begin{aligned} \Rightarrow \text{Info Age}(D) &= \frac{5}{14} \times 0.971 + \frac{4}{14} \times 0 + \frac{5}{14} \times 0.971 \\ &= 0.694 \end{aligned}$$

$$* \text{Gain}(\text{Age}) = \text{Info}(D) - \text{Info}_A(D)$$

$$\begin{aligned} &= 0.940 - 0.694 \\ &= 0.246 \end{aligned}$$

\* Information gain :-

|               |       |
|---------------|-------|
| Age           | 0.246 |
| Income        | 0.029 |
| Student       | 0.151 |
| Credit-Rating | 0.048 |

② Gain Ratio :

$$\Rightarrow \text{Gain Ratio}(A) = \frac{\text{Gain}(A)}{\text{split Info}(A)}$$

$$\Rightarrow \text{split Info}_A(D) = \sum_{j=1}^V \frac{|D_j|}{|D|} \times \log_2 \left( \frac{|D_j|}{|D|} \right)$$

\* For Attribute (Age) :-

$$\Rightarrow \text{Gain Ratio}(\text{Age}) = \frac{\text{Gain}(\text{Age})}{\text{split Info}(\text{Age})}$$

$$\begin{aligned} \Rightarrow \text{split Info}_{(\text{Age})} &= \frac{-5}{14} \log_2 \left( \frac{5}{14} \right) - \frac{4}{14} \left( \log_2 \left( \frac{4}{14} \right) \right) - \frac{5}{14} \log_2 \left( \frac{5}{14} \right) \\ &= \frac{-5}{14} (-1.485) - \frac{4}{14} (-1.807) - \frac{5}{14} (-1.485) \\ &= 0.530 + 0.516 + 0.530 \\ &= 1.576 \end{aligned}$$

$$\begin{aligned} \Rightarrow \text{Gain Ratio}_{(\text{Age})} &= \frac{\text{Gain}(\text{Age})}{\text{split Info}_{(\text{Age})}} \\ &= \frac{0.246}{1.576} \Rightarrow 0.156 \end{aligned}$$

$$\therefore \text{Gain Ratio}_{(\text{Age})} = 0.156$$



\* For Attribute (income) :-

| income | class |   | I(P,N) |
|--------|-------|---|--------|
|        | y     | N |        |
| high   | 2     | 2 | 1      |
| medium | 4     | 2 | 0.918  |
| low    | 3     | 1 | 0.811  |
|        | 9     | 5 | 0.940  |

$$\begin{aligned} \textcircled{i} \quad I(P,N) &\Rightarrow I(2,2) \Rightarrow -\frac{2}{4} \log_2\left(\frac{2}{4}\right) - \frac{2}{4} \log_2\left(\frac{2}{4}\right) \\ &\Rightarrow -\frac{2}{4} (-1) - \frac{2}{4} (-1) \\ &\Rightarrow 1 \end{aligned}$$

$$\begin{aligned} \textcircled{ii} \quad I(P,N) &\Rightarrow I(4,2) \Rightarrow -\frac{4}{6} \log_2\left(\frac{4}{6}\right) - \frac{2}{6} \log_2\left(\frac{2}{6}\right) \\ &\Rightarrow -\frac{4}{6} (0.585) - \frac{2}{6} (-1.585) \\ &\Rightarrow 0.918 \end{aligned}$$

$$\begin{aligned} \textcircled{iii} \quad I(P,N) &\Rightarrow I(3,1) \Rightarrow -\frac{3}{4} \log_2\left(\frac{3}{4}\right) - \frac{1}{4} \log_2\left(\frac{1}{4}\right) \quad (\text{iv}) \quad I(P,N) \Rightarrow (9,5) \\ &\Rightarrow -\frac{3}{4} (-0.415) - \frac{1}{4} (-2) \Rightarrow -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) \\ &\Rightarrow 0.811 \quad \Rightarrow 0.940 \end{aligned}$$

$$\begin{aligned} * \quad \text{Info}_{\text{Income}}(D) &= \sum_{j=1}^4 \frac{|D_j|}{|D|} \times \text{Info}(D_j) \\ &= \frac{4}{14} \times 1 + \frac{6}{14} \times 0.918 + \frac{4}{14} \times 0.811 \\ &= 0.911 \end{aligned}$$

$$* \text{Gain}(\text{Income}) = \text{Info}(D) - \text{Info}_A(D)$$

$$= 0.946 - 0.911$$

$$= 0.029$$

\* For Attribute (Income):

$$\Rightarrow \text{GainRatio}(\text{Income}) = \frac{\text{Gain}(\text{Income})}{\text{split info}(\text{Income})}$$

$$\Rightarrow \text{split info}(\text{Income}) = -\frac{4}{14} \log_2\left(\frac{4}{14}\right) - \frac{6}{14} \log_2\left(\frac{6}{14}\right) - \frac{4}{14} \log_2\left(\frac{4}{14}\right)$$

$$= -\frac{4}{14} (-1.807) - \frac{6}{14} (-1.222) - \frac{4}{14} (-1.807)$$

$$= 0.516 + 0.524 + 0.516$$

$$= 1.556$$

$$\Rightarrow \text{Gain Ratio}(\text{Income}) = \frac{\text{Gain}(\text{Income})}{\text{split info}(\text{Income})}$$

$$= \frac{0.029}{1.556}$$

$$= 0.0186 \approx 0.019$$

\* If an Attribute has maximum gain ratio then that attribute is taken to be the root node.

(03) Gini index :-

$$\Rightarrow \text{gini}(D) = 1 - \sum_{i=1}^n p_i^2$$

$$\Rightarrow \text{gini}_A(D) = \frac{|D_1|}{|D|} \text{gini}(D_1) + \frac{|D_2|}{|D|} \text{gini}(D_2)$$

\* For Attribute (Income):

$$\Rightarrow \text{gini} = 1 - \sum p_i^2 = 1 - \left[ \left(\frac{9}{14}\right)^2 + \left(\frac{5}{14}\right)^2 \right]$$

$$= 1 - [0.4132 + 0.1295]$$

$$= 1 - 0.5427$$

$$= 0.459$$

|       |    |    |
|-------|----|----|
| * L-4 | D1 | D2 |
| m-6   | LM | H  |
| H-4   | MH | L  |
|       | LH | M  |

$$\begin{aligned} * gini_A(D) &= \left[ \frac{10}{4} \left[ 1 - \left[ \left( \frac{6}{10} \right)^2 + \left( \frac{3}{10} \right)^2 \right] \right] + \frac{4}{14} \left[ 1 - \left[ \left( \frac{3}{4} \right)^2 + \left( \frac{1}{4} \right)^2 \right] \right] \right] \\ &= \frac{10}{4} (0.48) + \frac{4}{14} (0.375) \\ &= 0.343 + 0.107 \\ &= 0.450 \end{aligned}$$

$$\begin{aligned} * gini_A(D) &= \left[ \frac{10}{4} \left[ 1 - \left[ \left( \frac{7}{10} \right)^2 + \left( \frac{3}{10} \right)^2 \right] \right] + \frac{4}{14} \left[ 1 - \left[ \left( \frac{2}{4} \right)^2 + \left( \frac{2}{4} \right)^2 \right] \right] \right] \\ &= \frac{10}{4} [1 - 0.49 - 0.09] + \frac{4}{14} [1 - 0.25 - 0.25] \\ &= \frac{10}{4} \times 0.42 + \frac{4}{14} \times 0.50 \\ &= 0.306 + 0.143 \\ &= 0.433 \end{aligned}$$

$$\begin{aligned} * gini_A(D) &= \frac{8}{14} \left[ 1 - \left[ \left( \frac{5}{8} \right)^2 + \left( \frac{3}{8} \right)^2 \right] \right] + \frac{6}{14} \left[ 1 - \left[ \left( \frac{4}{6} \right)^2 + \left( \frac{2}{6} \right)^2 \right] \right] \\ &= \frac{8}{14} (0.469) + \frac{6}{14} (0.444) \\ &= 0.268 + 0.190 \\ &= 0.458 \end{aligned}$$

\* Gini index values:-

|    |   |       |
|----|---|-------|
| LM | H | 0.443 |
| MH | L | 0.450 |
| LH | M | 0.458 |



## Generation of confusion matrix

| Actual class   predicted class | Cancer = yes | cancer = no | Total |
|--------------------------------|--------------|-------------|-------|
| Cancer = yes                   | 90           | 210         | 300   |
| cancer = no                    | 140          | 9560        | 9700  |
| Total                          | 230          | 9770        | 10000 |

| A/P      | C  | $\neg C$ |     |
|----------|----|----------|-----|
| C        | TP | FN       | P   |
| $\neg C$ | FP | TN       | N   |
|          | P' | N'       | All |

(I) precision :-  $\frac{TP}{TP+FP}$

$$= \frac{90}{90+210} = \frac{90}{300} = 0.39$$

$\Rightarrow 39\%$

(II) specificity :-  $\frac{TN}{FP+TN}$

$$= \frac{9560}{140+9560} = 98.56\%$$

(III) Accuracy :-

$$\frac{TP+TN}{TP+FP+FN+TN} \Rightarrow \frac{90+9560}{90+140+9560+210}$$

$$= 0.965$$

$$\text{Precision} \Rightarrow \frac{TP}{TP+FP} \Rightarrow \frac{90}{90+140} \Rightarrow 0.3913 \Rightarrow 39.13\%$$

$$\text{Recall} \Rightarrow \frac{TP}{TP+FN} \Rightarrow \frac{90}{90+210} \Rightarrow 0.3$$

$$\text{F1-Score} \Rightarrow \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = \frac{2 \times 0.39 \times 0.3}{0.39 + 0.3}$$

$$\Rightarrow \frac{0.23478}{0.6913}$$

$$\text{error rate} \Rightarrow 1 - \text{Accuracy} \Rightarrow 33.96\%$$

$$= 1 - 0.965$$

$$= 0.035$$

$$= 3.5$$

